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INFLUENCE OF EDUCATION ON FIREARM MORTALITY

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Problem Statement

Seriousness of the firearm Mortality

- Firearm mortality remain a significant public health concern [1].
- In the USA especially, gun death rates have affected the public health challenges [1].

Overlooked Factor

- However, research and public discourse have overlooked how educational attainment influences gun mortality risk.
- Most existing studies focus on race, age, rurality and gun ownership rates [2].
- Education level is rarely treated as the major factor [2].

Gap In the analysis

- According to Jha et. al. [3], the contribution of education to firearm mortality has yet to be studied although variations in education has been registered previously

Recent links

- Hamilton and Kposowa [4] found that higher education levels are associated with lower rates of firearm suicide and total gun deaths.

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ABOUT DATASET

- To address the lack of focus on education in gun death research and to directly compare education with firearm deaths, the dataset “Gun Deaths in America-CDC” [5] was used.

- Following are the reasons for choosing the dataset:

Reasons	Description
Availability	Dataset is publicly available on Kaggle.
Cost	Free to access and use.
Domain Relevance	Covers Education and Firearm Mortality which facilitates a direct link between them.
Comparability	Allows direct comparison with previous studies that claims education level also affects firearm mortality rate.
Data Quality	Sourced from the CDC, ensuring high credibility.

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DATASET PROPERTIES

- Information about gun-related deaths in the United States.
- Total number of rows: **100798**
- Total number of columns: **9**

ATTRIBUTES:

- Year of Death
- Month of Death
- Intent
- Police Involvement
- Gender Of The Victim
- Age Of The Victim
- Race Of The Victim
- Place Of Shooting
- Education Level

INTENT OF DEATH:

- Suicide
- Homicide
- Accidental
- Undetermined

EDUCATION LEVEL

- Less Than High School
- High School/General Educational Development
- Some College
- BA+

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METHODOLOGY

Data Source:
Kaggle
CSV Dataset

→

HDFS Ingestion:
Hdfs dfs –put
/user/Hadoop/myda
taset

→

Hive:
Create Database
Create Table

→

HiveQL Analysis:
COUNT(*)
GROUP BY
WHERE IN

→

Visualization:
Apache Zeppelin
Graph Chart
Line Chart

Download Dataset from Kaggle

↓

Upload CSV file to HDFS using PuTTY

↓

Create Hive Database and Table

↓

Execute HiveQL Queries

↓

Visualize in Apache Zeppelin

↓

Analyze Education Levels Against Firearm Mortality Rates

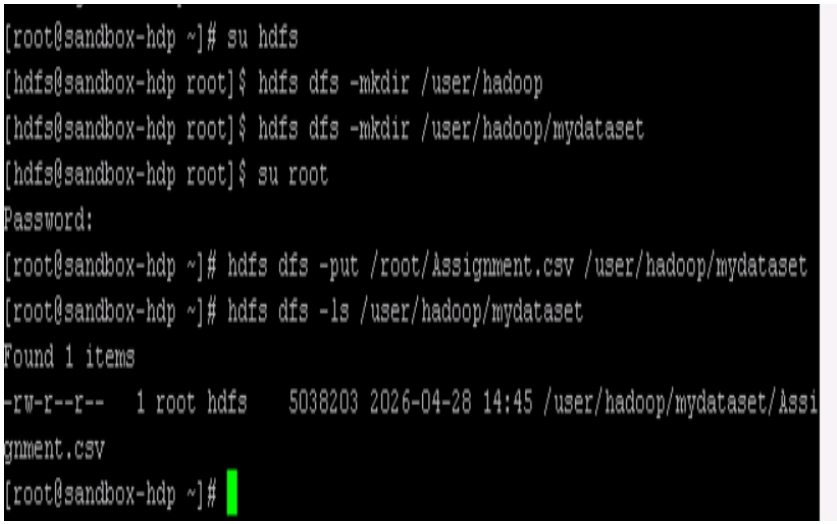


Fig. 1: Uploading dataset to HDFS using PuTTY

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STORAGE LAYER (HDFS)

- Dataset uploaded from local machine to Hadoop Distributed File System using PuTTY.
- Copied the dataset from local machine using the path of the file.
- Created directory as in Figure 1. to add the dataset in HDFS.

PuTTY is a free open-source terminal emulator and SSH client [6]. It was used to:

- Connect to Hadoop environment.
- Upload the dataset into HDFS.
- Run Hive queries from command line.

Hadoop Distributed File System?

- HDFS is a distributed storage system that stores very large datasets across multiple machines and ensures high data access speed [7].
- HDFS was used for its:
- **High Throughput:** Optimized for batch processing jobs.
- **Scalability:** Stores large volumes of data across distributed nodes.
- **Fault Tolerance**: Data is replicated across nodes. Facilitates reliability in case of failures.
- **Ecosystem Integration:** Easily integrates with Hive, which was used for querying and processing.
- **Not Schema Based**: Data of any type can be stored.

Hadoop Over Other Systems

- **GlusterFS:** Works better as a general-purpose file server than dedicated storage layer in Hadoop.
- **QFS:** Less commonly used and has smaller ecosystem than HDFS.
- **CephFS:** Adds more complexity as it requires configuring multiple components.
- **Azure Data Lake:** Requires extra login setup and adds more complexity.
- **Amazon S3:** S3 is object storage and does not support true file system operations.
- **Google Cloud Storage:** Requires cloud dependency and external login setup.

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DATA PROCESSING WITH HIVE

Steps Performed:

1. Created a database named cw2.
2. Defined a table “gun_deaths” inside the database, matching the 9 columns of the CDC dataset .
3. Loaded the data from the dataset into the Hive table.
4. Used SQL to explore the dataset. Ran *SELECT * FROM gun_deaths LIMIT 5;* to check column values and confirm data loaded correctly.
5. Performed queries to identify the education level with the highest number of deaths using *GROUP BY education ORDER BY deaths DESC LIMIT 1* as shown in figure 3.
6. Counted number of gun deaths by suicide and homicide using *COUNT(*)* with *WHERE intent IN (“Suicide”, ”Homicide”).*
7. Performed additional queries like SELECT education, COUNT(*) AS death FROM gun_deaths GROUP BY education to analyze deaths across different educational levels.

Why Hive?

- Hive runs on top of HDFS and hence data stays in distributed storage.
- Hive handles large rows smoothly and without performance issue.
- Hive is also good at batch processing which resonates with this workload.
- Hive uses SQL. There is no need to learn a new query language unlike Spark and Pig, which require more procedural code.
- Hive integrates well with Zeppelin and BI tools while Pig has smaller ecosystem support. Spark needs additional Spark session configuration or cluster resource tuning.

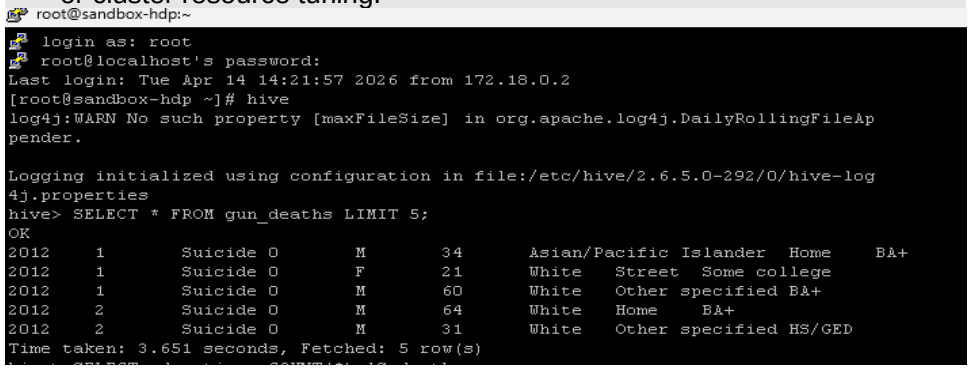


Fig. 2: Query that shows first five rows of the table

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DATA PROCESSING WITH HIVE

```
hive> SELECT education, COUNT(*) AS total_deaths
    > FROM gun_deaths
    > GROUP BY education
    > ORDER BY total_deaths DESC
    > LIMIT 1;
Query ID = root_20260414144402_d73635c9-b090-4113-b83c-6fcfda89c133
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1776169531308_0034)

--------------------------------------------------------------------------------
        VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
--------------------------------------------------------------------------------
Map 1 ..........   SUCCEEDED      1          1        0        0       0       0
Reducer 2 ......   SUCCEEDED      1          1        0        0       0       0
Reducer 3 ......   SUCCEEDED      1          1        0        0       0       0
--------------------------------------------------------------------------------
VERTICES: 03/03  [==========================>>] 100%  ELAPSED TIME: 6.66 s
--------------------------------------------------------------------------------
OK
HS/GED  42927
Time taken: 11.99 seconds, Fetched: 1 row(s)
```

Fig. 3: Query that finds the education level with the highest total gun deaths.

```
hive> SELECT intent, COUNT(*) AS deaths
    > FROM gun_deaths
    > WHERE intent IN('Suicide','Homicide')
    > GROUP BY intent;
Query ID = root_20260414143518_8defcfb4-9e7f-4d91-9eb0-48790753d382
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1776169531308_0033)

--------------------------------------------------------------------------------
        VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
--------------------------------------------------------------------------------
Map 1 ..........   SUCCEEDED      1          1        0        0       0       0
Reducer 2 ......   SUCCEEDED      1          1        0        0       0       0
--------------------------------------------------------------------------------
VERTICES: 02/02  [==========================>>] 100%  ELAPSED TIME: 5.79 s
--------------------------------------------------------------------------------
OK
Homicide        35176
Suicide 63175
Time taken: 6.684 seconds, Fetched: 2 row(s)
```

Fig. 4:Query that finds the total gun death counts for Suicide and Homicide

```
hive> SELECT education, COUNT(*) AS death
    > FROM gun_deaths
    > GROUP BY education;
Query ID = root_20260414143419_ba0b3191-f994-4924-a62a-424df6d25d4e
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1776169531308_0033)

--------------------------------------------------------------------------------
        VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
--------------------------------------------------------------------------------
Map 1 ..........   SUCCEEDED      1          1        0        0       0       0
Reducer 2 ......   SUCCEEDED      1          1        0        0       0       0
--------------------------------------------------------------------------------
VERTICES: 02/02  [==========================>>] 100%  ELAPSED TIME: 7.21 s
--------------------------------------------------------------------------------
OK
        1422
BA+     12946
HS/GED  42927
Less than HS    21823
Some college    21680
Time taken: 9.741 seconds, Fetched: 5 row(s)
```

Fig. 5: Query that finds the total gun death count for each education level

- Figure 2 showed the first five rows of the table and confirmed that the dataset has been successfully loaded into the Hive table.
- Figure 3 showed that the education level HS/GED had the highest total gun deaths.
- Figure 4 showed that the total number of suicides was 63,175 and the total number of homicides was 35,176.
- Figure 5 showed the total gun death count for each education level. The output revealed that the HS/GED had highest total gun deaths, followed by Less than HS and Some college. The gun deaths among BA+ were comparatively fewer. Few deaths with no education label appeared to be uncategorized.

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DATA VISUALIZATION

Steps Performed:

- Queries were performed on Apache Zeppelin to obtain the graphical representations.
- “%jdbc” was used as interpreter prefix to execute queries for Hive databases.
- Results presented using Line graphs, Pie charts and Bar charts.
- The link between education levels and intent of deaths were visualized and examined in the data visualization phase.

Zeppelin

- Apache Zeppelin is an open-source web-based notebook tool for data visualization and data analysis [8] [10].
- Apache Zeppelin provides notebook-style interface to run queries and visualize results [8].
- It works smoothly with Big Data tools (Hive).
- It helps in automatic generation of charts for analysis of the data.
- It has interactive and user-friendly interface.
- It allows iterative querying and refinement.

Why Zeppelin?

- Hadoop cluster already includes Zeppelin and Hive integration.
- Zeppelin supports iterative analysis whereas BI tools are more dashboard-focused [9].
- BI platforms are designed for dashboarding and require data extraction. Zeppelin executes Hive queries directly on data stored in HDFS.
- Tableau also does not directly operate on HDFS efficiently.

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INSIGHTS FROM THE DATA

- The relationship between gun deaths and education level was obtained using SQL queries. Following insights were obtained:

```
%jdbc(hive)
SELECT intent,
       COUNT(*) AS total_deaths
FROM gun_deaths
GROUP BY intent
ORDER BY total_deaths DESC;
```

FINISHED

settings

intent	total_deaths
Suicide	63175
Homicide	35176
Accidental	1639
Undetermined	807
	1
intent	1

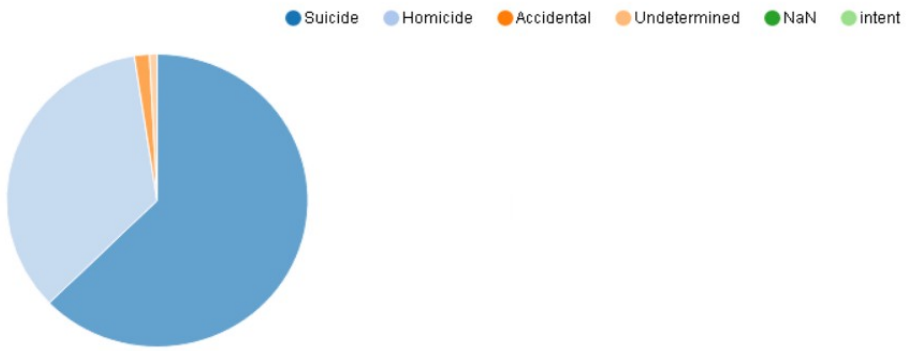


Fig. 6: Count Of Gun Deaths With Respect To Intent

- Suicide accounts for the largest cause of gun deaths with 63,175.
- Homicide is the second most common intent for gun deaths with 35,176.
- Accidental and undetermined gun deaths are rare.
- It was inferred that Suicide accounts for nearly two-thirds of all gun deaths.

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INSIGHTS FROM THE DATA

```
%jdbc(hive)
SELECT education,
       COUNT(*) as total_deaths
FROM gun_deaths
WHERE education IS NOT NULL AND education!=''
GROUP BY education
ORDER BY total_deaths DESC
```

settings

education	total_deaths
HS/GED	42927
Less than HS	21823
Some college	21680
BA+	12946
education	1

FINISHED

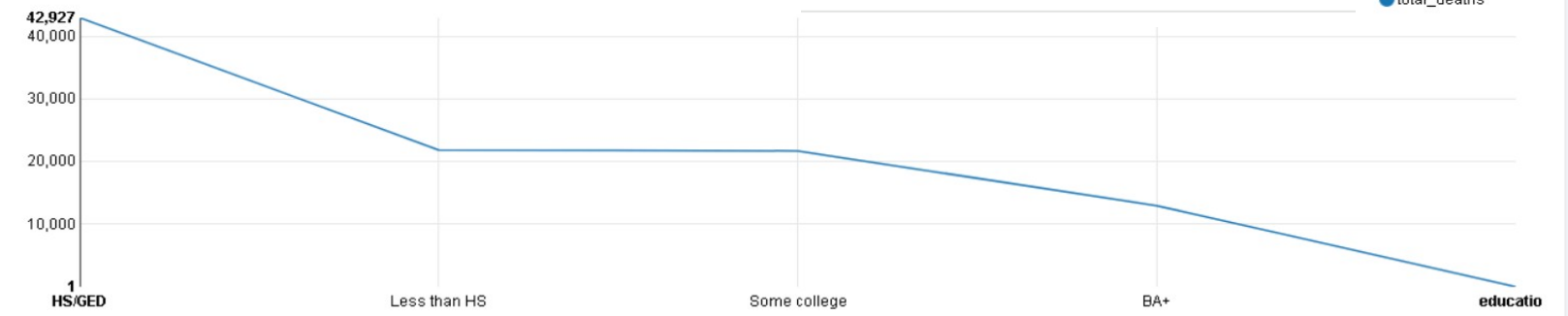


Fig. 7: Number Of Gun Deaths By Education Level Highest To Lowest

- HS/GED has the highest total gun deaths, nearly double the second highest demographic. This may be due to higher firearm ownership and greater exposure to high-risk environments.
- Less than HS and Some college are nearly identical with only 143 deaths apart. The small gap suggests that high school completion alone is not a clear dividing line.
- BA+ has the lowest total gun deaths. This may be due to better employability and safer neighborhoods.
- Those with an education level lower than BA+ appear to be at greater risk for gun-related deaths.
- It was inferred that higher education level is associated with fewer total gun deaths.

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INSIGHTS FROM THE DATA

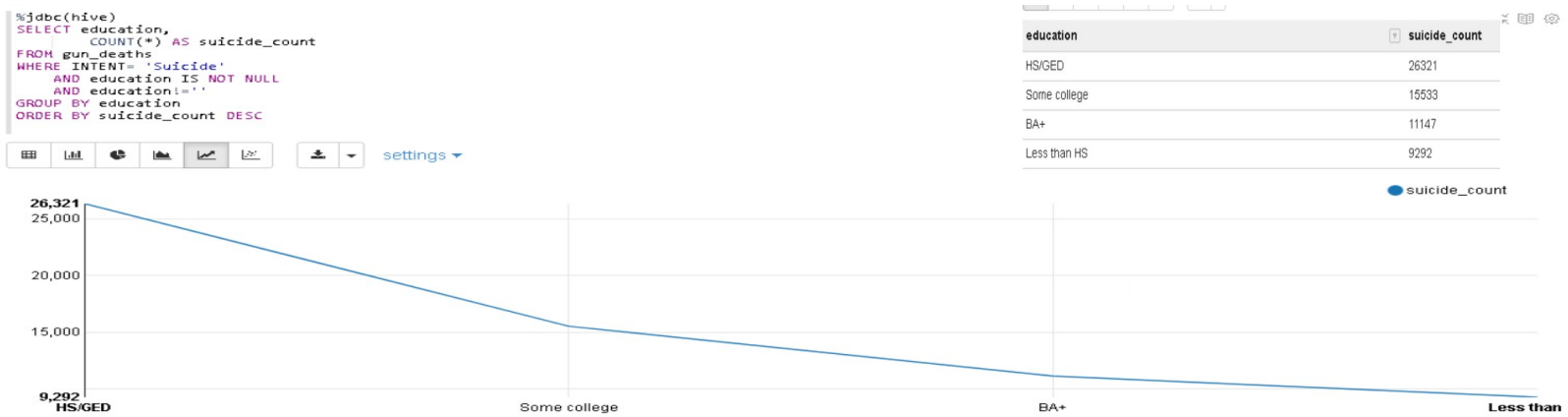


Fig. 8: Suicide Counts With Respect To Education Level

- HS/GED has the highest suicide count by far, followed by Some college, then BA+.
- Less than HS has the lowest suicide count.
- Unlike total gun deaths, higher education does not mean fewer suicides.
- Those with an education level BA+ are still susceptible to firearm deaths, mostly by suicide. This may be due to mental health issue that are not solely based on education level.
- However, their total number of deaths by suicide is still much lower than the HS/GED education level.

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INSIGHTS FROM THE DATA

```
%jdbc(hive)
SELECT
    education,
    COUNT(*) AS homicide_deaths
FROM gun_deaths
WHERE intent = 'Homicide'
  AND education IS NOT NULL
  AND education != 'NaN'
  AND education != ''
GROUP BY education
ORDER BY homicide_deaths DESC;
```

education	homicide_deaths
HS/GED	15649
Less than HS	11839
Some college	5640
BA+	1559

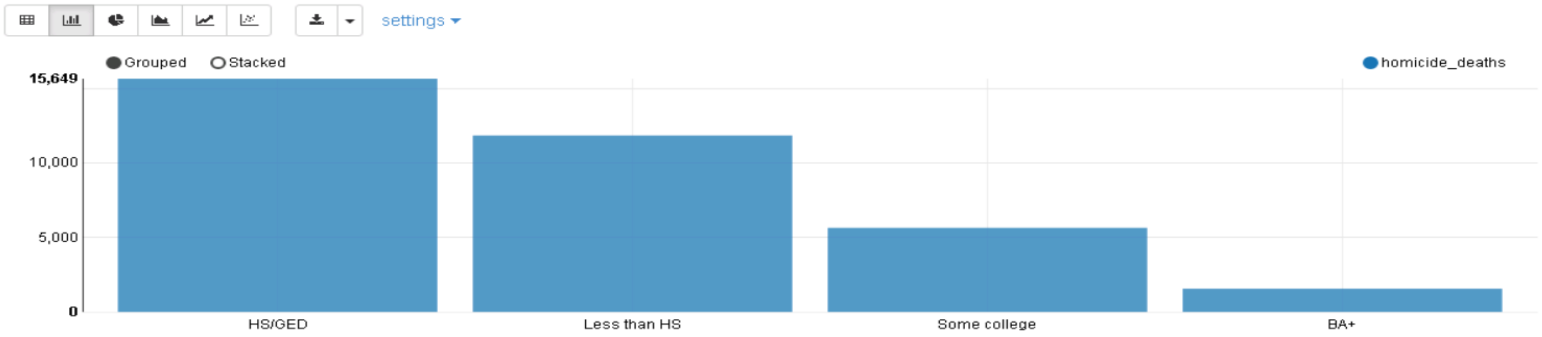


Fig. 9: Homicide Count With Respect To Education Level

- HS/GED has the highest homicide count .
- BA+ has 90% fewer homicides than HS/GED.
- Clearly shown that higher education = fewer homicide deaths.
- Less than HS is the only group where homicide exceeds suicide.
- Education affects suicide and homicide differently: homicide drops sharply whereas suicide drops only modestly.
- It can be inferred that higher education strongly protects against homicide but only weakly protects against suicide.

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CONCLUDING REMARKS

- There appears to be a significant link between education and gun deaths.
- Lower levels of educational attainment are associated with higher gun death rates.
- Individuals in the HS/GED group was found to have highest number of firearm deaths.
- Gun deaths appeared most prevalent in suicide cases.
- Homicide is the second most common cause of firearm deaths.
- The analysis showed that higher-educated groups are more susceptible to firearm death related to suicide than homicide.

- Homicide was more prevalent in lower-educated groups. It was fewer among the BA+ education level demographics.
- The data lake architecture scaled effectively for 100798 records without performance issues.
- It indicated that the chosen system was highly suitable for handling this dataset.
- It also demonstrated that the solution can support larger datasets and future expansion.

- Although the analysis showed important links, the education level categories may hide socioeconomic factors.
- The dataset does not consider variables such as mental health history or economic conditions.
- The data is limited and findings may not generalize to other populations or geographic regions.
- In future, the data may be expanded to include additional years, states and more records to populate the dataset further.
- The attributes such as gun ownership rates, mental health resources, employability and poverty rates may be added.
- The queries may also be performed to investigate other relationships such as education and age, education and gender and death and location of death for clearer insights.

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